How Transactions are Verified and Added to the Mempool

A **full node** checks all transactions to make sure they are valid and have not been used before. If a transaction is good, the node will verify it and add it to it's memory pool. Then it will share it with other nodes to double-check. Finally, if the majority agree, it will be taken out of everyone's mempool to become permanent part of the *blockchain*.

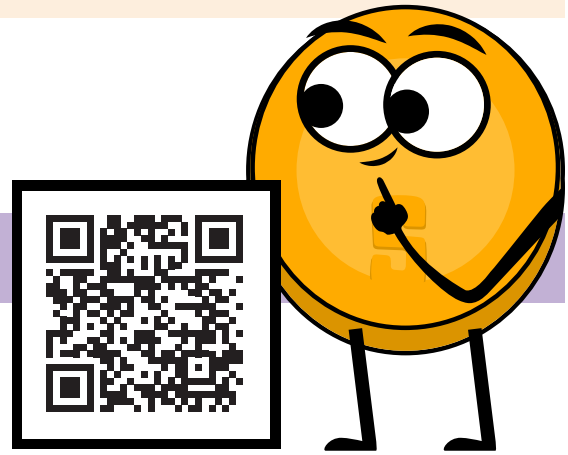
Transactions in the ***Bitcoin Network*** are taken out of the mempool and confirmed when they are included in a block, which is then added to the *blockchain*. However, there are several reasons why a transaction might not be confirmed after 72 hours:

1. Low fee: Transactions with a low fee may not be processed quickly enough, as miners are more likely to choose transactions with higher fees to include in their blocks.
2. Network congestion: If the network is congested, there may be a delay in confirming transactions, even if they have a high fee.
3. Double spend attempt: If a malicious actor attempts to double spend, their transaction may be rejected by the network.
4. Incorrect or incomplete data: If a transaction contains incorrect or incomplete data, it may be rejected by the network.
5. Malformed transaction: If a transaction is malformed, it may be rejected by the network.

To avoid having transactions rejected, it's recommended to include a fee that is high enough to ensure the transaction is processed in a timely manner, and to double check that all the data in the transaction is correct before sending it.

A **DDoS** (Distributed Denial of Service) attack is an attempt to make the network unavailable to users by overwhelming it with too much traffic from multiple sources. This attack aims to disrupt normal traffic of a website or service by overwhelming it with fake traffic, making it difficult or impossible for real users to access it.

7.3.1 Class Exercise: On Hold: Examining the Unconfirmed Transactions of the Bitcoin Network



Class Exercise. Follow the following instructions:

1. Visit the website *https://bits.monospace.live/*